

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCC3073 / UCCG3073 DATA SCIENCE

WEDNESDAY, 17 MAY 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF SCIENCE (HONOURS)
STATISTICAL COMPUTING AND OPERATIONAL RESEARCH

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the **FIRST (1)** question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

UCCC3073 / UCCG3073 DATA SCIENCE**SECTION A (Answer ALL questions)**

- Q1. (a) With the growth of digitalized business models, more data are collected to maximise the gain from the business models.

Consider Netflix media company:

- (i) Identify and explain any TWO (2) characteristics of big data that are important for Netflix. In your answer, provide an example for each of the identified characteristics in the context of Netflix. (6 marks)
- (ii) Provide any TWO (2) examples of data that should be collected by Netflix to perform big data analysis. (2 marks)
- (iii) Describe any TWO (2) services that can be offered by big data analysis. (2 marks)
- (iv) Describe any TWO (2) benefits that can be gained by Netflix for having such big data analysis. (2 marks)

- (b) Consider the following data.

17, 99, 19, 88, 90, 79, 85, 70

Normalise the data using the following methods. Round your answer to 2 decimal points.

- (i) z-score. (4 marks)
- (ii) min-max. (3 marks)

Q1. (Continued)

- (c) Department of Statistics Malaysia conducted a research study on the residency population in Malaysia from year 2010 to 2019. Table 1.1 show the population summary (in percent and in 1 millionth) of Malaysia citizen categorized by age group.

Table 1.1

Population (%)	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
0-19	39.25	38.56	37.91	37.26	36.6	36.01	35.48	34.93	34.43	33.93
20-39	31.62	31.85	32.07	32.32	32.61	32.84	33.00	33.18	33.32	33.4
40-59	20.82	21	21.15	21.26	21.34	21.39	21.43	21.45	21.47	21.53
60-79	7.42	7.68	7.95	8.22	8.49	8.76	9.06	9.36	9.65	9.96
80+	0.89	0.91	0.92	0.94	0.96	1.00	1.03	1.08	1.13	1.18
Population (in 1 millionth)	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Total	26.26	26.62	26.96	27.33	27.70	28.06	28.40	28.74	29.06	29.38

- (i) Identify sample and population of the study. (2 marks)
- (ii) Derive FOUR (4) insights from the study. (4 marks)
- [Total : 25 marks]

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- Q2. (a) Given a population mean and population sample standard deviation of 20 paired differences are 6 and 4.8, a study was performed to study the differences between results derived from population and from sample, with the sample mean of 5.7. A hypothesis testing is carried out for the following:

H_0 = There is no difference on the results derived from population and sample.

H_a = Results derived from population are different from results derived from sample.

Using a 95% confidence interval for 2-tail normal distribution, determine the following and round your answer to 2 decimal points.

- (i) Standard error. (1 mark)
 - (ii) z test. (1 mark)
 - (iii) p-value. (2 marks)
 - (iv) upper- and lower-bounds. (3 marks)
- (b) Using the answers of Q2(a), justify whether the null hypothesis can be rejected. (2 marks)
- (c) Using the answer of Q2(a), justify whether the outlier present in the data. (2 marks)
- (d) Describe FOUR (4) possible outcomes from hypothesis testing. (4 marks)
- (e) Given a bag of 20 balls of which 6 are red and the remaining are blue. Three (3) balls were drawn from the bag and did not put back. Draw the probability tree to determine the probability that all the drew balls are blue. (10 marks)
- [Total : 25 marks]

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Q3. Consider the following transactional data from a computer shop.

Table 3.1

Transaction ID	Mouse	HDMI	VGA	Keyboard
T1	1	1	1	1
T2	1	0	1	1
T3	0	0	1	1
T4	0	1	0	0
T5	1	0	1	1
T6	1	1	0	1

Assume that the minimum support is 50% and confidence is 80%.

- (a) Find all the frequent item sets. In your answer, show the candidate item sets obtained at each step of the pruning stage and list down all the frequent item sets. (10 marks)
 - (b) Generate the Apriori rule(s) for frequent item set L3. In your answer, show the working steps on how you obtain the rules and list down the finalised rules. (8 marks)
 - (c) Determine support and lift for the identified rule(s) in Q3(b). (2 marks)
 - (d) Identify the least popular item that you observe from the Apriori rule(s) in Q3(b). (1 mark)
 - (e) Based on your observations in Q3(d), provide TWO (2) suggestions to the retailer to improve the sales. (4 marks)
- [Total : 25 marks]

UCCC3073 / UCCG3073 DATA SCIENCE**SECTION B (Choose only ONE question)**

- Q4. Table 4.1 shows the confusion table for the naïve bayes (NB) models with and without misclassification cost applied. Model 1 is the NB with equal cost. Model 2 is the NB with unequal cost of which the cost for false positive $Cost_{FP}$ is 20 and the cost for true positive, $Cost_{TP}$ is -10.

Table 4.1

N = 5,000		Prediction values			
		Model 1		Model 2	
		No	Yes	No	Yes
Actual values	No	1000	0	900	100
	Yes	3000	1000	800	3200

- (a) Calculate the baseline performance. (2 marks)
- (b) Calculate the following evaluation measures for both models. For each measure, identify the best model. Round your answer to the nearest 2 decimal points.
- (i) Misclassification rate. (3 marks)
 - (ii) Recall. (3 marks)
 - (iii) F_1 score. (3 marks)
 - (iv) Profit cost per record. (3 marks)
- (c) Using the computed results in Q4(a) and Q4(b), justify whether you would propose the model to the company. If yes, identify which model you would propose. (7 marks)
- (d) Provide ONE (1) business example that you would propose the model in Q4(c) and justify how would the model benefits the business. (2 marks)
- (e) Provide ONE (1) business example that you would not propose the model in Q4(c) and justify why such model should not be considered. (2 marks)
- [Total : 25 marks]

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- Q5. You are required to propose a diagnostic model for a hospital to screen cancer patients. Due to a lack of book-keeping system in the hospital, you are given only 60 patient samples to build the diagnostic model. You decided to use 50 samples for training the model and the remaining 10 samples for testing the model. In the model development, you constructed 2 models, namely Naïve Bayes (NB) and K nearest neighbour (KNN). Table 5.1 is the prediction results of the testing set made by NB and KNN.

Table 5.1

Patient Id	Actual diagnosis	NB Prediction	KNN Prediction
P1	Healthy	Healthy	Cancer
P2	Healthy	Healthy	Healthy
P3	Healthy	Healthy	Healthy
P4	Healthy	Healthy	Healthy
P5	Healthy	Healthy	Cancer
P6	Cancer	Cancer	Cancer
P7	Cancer	Cancer	Healthy
P8	Cancer	Cancer	Healthy
P9	Cancer	Healthy	Cancer
P10	Cancer	Cancer	Cancer

- (a) Compute the confusion matrix for both the NB and KNN models. (4 marks)
- (b) Calculate the following evaluation measures for both models. For each measure, identify the best model. Round your answer to the nearest 4 decimal points.
- (i) Classification accuracy. (3 marks)
 - (ii) Recall. (3 marks)
 - (iii) Specificity. (3 marks)
 - (iv) False discovery rate. (3 marks)
- (c) Using the computed results in Q5^(a) and Q5^(b), justify which model is a better fit for the data. (7 marks)
- (d) Justify whether you would recommend the model identified in Q5^(c) to the hospital. (2 marks)

[Total : 25 marks]

APPENDIX

Regression	
Straight line, $\hat{y} = \beta_0 + \beta_1 x$	t value, $t = \frac{\hat{\beta}}{SE}$
Coefficients $\hat{\beta}_1 = \frac{p \sum xy - \sum x \sum y}{p \sum x^2 - (\sum x)^2}$	Standard errors of coefficients $SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$
$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$,	$SE(\hat{\beta}_0) = s \sqrt{\frac{1}{p} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$
Coefficient of Determination R^2 $= \left[\frac{p \sum xy - \sum x \sum y}{\sqrt{(p \sum x^2 - (\sum x)^2) * (p \sum y^2 - (\sum y)^2)}} \right]^2$	Residuals $\hat{\epsilon} = y - \hat{y}$ $s^2 = \frac{\sum \hat{\epsilon}^2}{p - 2}$
$R = \sqrt{R^2}$	$s = \sqrt{s^2}$

Naïve Bayes
Bayes' rule, $P(H E) = \frac{P(E H)P(H)}{P(E)}$
Laplace correction, $\frac{x_i + 1}{c + \alpha}$
Probability density function for a Gaussian distribution, $pdf(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\bar{x})^2}{2\sigma^2}}$

Association Rules
Association rule, $x \rightarrow y$
Support, $s = \frac{freq(X,Y)}{N}$
Confidence, $c = \frac{freq(X,Y)}{freq(X)}$
Lift, $l = \frac{Confidence(x,y)}{Support(y)}$

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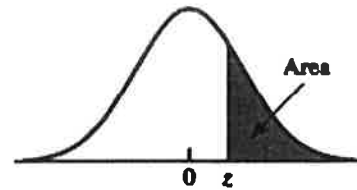
Evaluation metrics	
Classification accuracy, $acc = \frac{TN+TP}{TN+TP+FN+FP}$	Precision, $PPV = \frac{TP}{TP+FP} = 1 - FDR$
Misclassification rate, $1 - acc$	F1 score, $F_1 = \frac{2TP}{2TP+FP+FN}$
Sensitivity / Recall, $recall = \frac{TP}{TP+FN}$	Overall model cost $= TN.Cost_{TN} + FP.Cost_{FP} + FN.Cost_{FN} + TP.Cost_{TP}$
Specificity, $spec = \frac{TN}{TN+FP}$	Model cost per case $= \frac{\text{Overall model cost}}{N}$
False discovery rate, $FDR = \frac{FP}{FP+TP}$	Model profit per case $= -\text{model cost per case}$

Proximity measures	
Euclidean distance, $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$	Hamming distance, $d(a, b) = (a \oplus b)$
Manhattan distance, $d(x, y) = \sum_{i=1}^n x_i - y_i $	Simple matching coefficient, $SMC = \frac{\text{num of matching attribute values}}{\text{num of attributes}}$
Supremum distance, $d(x, y) = \max(x_i - x , y_i - y)$	Jaccard coefficient, $J = \frac{\text{num of matching presences}}{\text{num of attributes not involved in 00 matches}}$
	Cosine similarity, $\cos(x, y) = \frac{x \cdot y}{ x y }$

Miscellaneous	
Min-Max normalisation, $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$	Confidence intervals, $CI = \bar{x} \pm z * SE$
z-score normalisation, $x' = \frac{x - \mu}{\sigma}$	Standard error, $SE = \frac{\sigma}{\sqrt{n}}$
Mean, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Z-test, $Z = \frac{\bar{x} - \mu}{SE_{\bar{x}}}$
Sample standard deviation, $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$	

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Table 4 Normal Curve Areas
 Standard normal probability in right-hand tail
 (for negative values of z , areas are found by symmetry)



z	Second decimal place of z									
	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0352	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.00135									
3.5	.000 233									
4.0	.000 031 7									
4.5	.000 003 40									
5.0	.000 000 287									

From R. E. Walpole, *Introduction to Statistics* (New York: Macmillan, 1968).